

GOODNESS OF FIT OF THE GENERALIZED EXTREME VALUE DISTRIBUTION BASED ON THE KULLBACK-LEIBLER INFORMATION

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Summary: The goodness of fit of the Generalized Extreme Value (GEV) distribution is considered using the Kullback-Leibler (KL) information as well as a new proposed measure. Since the class of GEV distributions covers a wide class of distributions such as the normal, log Weibull, log Gamma and exponential for instance, this goodness of fit procedure can also be regarded as a model selection procedure among the class of GEV distributions. The new proposed measure is found to be easier to use than the KL information.

1. INTRODUCTION

In De Waal, Groenewald and Kemp (1995) the effect of perturbing the normal to the class of Generalized Extreme Value distributions is considered. Various measures to quantify the effect of perturbation have been considered such as the Kullback-Leibler measure (Geisser, 1992, Johnson and Geisser, 1982) of divergence of a GEV distribution from the normal. All these measures are found to be decreasing functions of the perturbation parameter k in the class of GEV distributions for the examples considered and maximum perturbation occurs when $k \rightarrow 0$. If $k \rightarrow \infty$ the GEV distribution becomes the normal.

It becomes a matter of interest to fit such a GEV distribution to a set of data and consider the goodness of fit. Various statistics for testing goodness of fit or selecting the appropriate model can be found in the literature (see

section 3). The Kullback-Leibler measure of information (Kullback, 1959) is a well-known measure to discriminate between two distributions and we will pay special attention to this measure in this paper. We will especially consider the procedure proposed by Ebrahimi, Habibullah and Soofi (1992) where they use the KL measure to propose a test for testing exponentiality. Our aim is to fit a GEV distribution to data, which usually are lifetime data, by selecting the parameter k of the GEV distribution using the KL information. A new measure is also proposed and a comparison between the two has been done. We did not attempt to make a comparison of these methods with the Bayes factor, because we have a diffuse prior on k and some people regard them as a little liberal in such a case. We, however, still intend to make such a comparison in future communication.

2. THE KULLBACK-LEIBLER MEASURE OF DISCRIMINATION

Consider a random sample $X_1, \dots, X_n$ from a probability distribution $F(x)$ with density function $f(x)$ over a non-negative support. Suppose we are interested to know if the sample comes from a specified probability distribution $F_0(x)$ with density function $f_0(x)$, then the KL information measure on the divergence between $F(x)$ and $F_0(x)$ is defined as (Kullback (1959))

$$(1) \quad I(F, F_0) = \int_0^{\infty} f(x) \ln \left[\frac{f(x)}{f_0(x)} \right] dx.$$

The entropy of the distribution $F(x)$ is defined as

$$(2) \quad H(F) = - \int_0^{\infty} f(x) \ln f(x) dx$$

and is estimated from the sample by (Vasicek (1976))

$$(3) \quad \hat{H}_{mn} = \frac{1}{n} \sum_{i=1}^n \ln \left\{ \frac{n}{2m} \left[X_{(i+m)} - X_{(i-m)} \right] \right\}$$

where the window size m is a positive integer smaller than $\frac{n}{2}$, $X_{(j)} = X_{(1)}$ if $j < 1$, and $X_{(j)} = X_{(n)}$ if $j > n$, $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ being the order

statistics of the sample.

Suppose $F_0(x)$ is a Gamma $(k, 1)$ distribution with density function

$$f_0(x) = \frac{1}{\Gamma(k)} e^{-x} x^{k-1}, \quad x > 0,$$

then

$$(4) \quad \int_0^{\infty} f(x) \ell n f_0(x) dx = -\ell n \Gamma(k) - E(X) + (k-1)E(\ell n X).$$

An estimate for $I(F, F_0)$ from the sample is therefore

$$\hat{I}_{mn} = -\hat{H}_{mn} + \ell n \Gamma(k) + \bar{X} - (k-1)\overline{\ell n X}$$

where $\bar{X} = \frac{1}{n} \sum X_i$

and $\overline{\ell n X} = \frac{1}{n} \sum \ell n X_i$.

It is known that $I(F, F_0) \geq 0$ and if the sample is drawn from a Gamma $(k, 1)$ population, then $I(F, F_0) = 0$. Large values of $\hat{I}_{mn}$ indicate that the sample is not from F_0 , which means, in the above case, the sample is not from a Gamma $(k, 1)$ distribution. Using the same monotone transformation as used by Ebrahimi et al. (1992), namely

$$(5) \quad \widehat{KL}_{mn} = \exp(-\hat{I}_{mn}), \quad 0 < \widehat{KL}_{mn} < 1$$

we can judge from $\widehat{KL}_{mn}$ if the sample is from a Gamma $(k, 1)$ distribution.

Small values of $\widehat{KL}_{mn}$ indicate that the sample is not from Gamma $(k, 1)$.

In fact, we can calculate from a Monte Carlo simulation $P(\widehat{KL}_{mn} \leq kl_{mn} | F_0)$

and judge accordingly, where kl is the calculated value from the sample.

Example 1

A sample of $n = 66$ has been drawn. It is calculated that $\bar{x} = 3.9752$, $\overline{\ell n x} = 1.2561$, $\hat{H}_{mn} = 1.9073$ and $\widehat{KL}_{mn} = 0.9146$. We conjecture that the sample has been drawn from a Gamma $(k = 4, 1)$ distribution. The window size m is taken as 9. From table 1 of Ebrahimi et al. (1992) we obtain a window size 7 for $n = 66$ if the sample is from an exponential distribution.

We found from a number of simulations that roughly we can work on a window size $m = m^* + k - 2$ for the Gamma $(k, 1)$ distribution, where m^* is the tabulated value in Ebrahimi *et al.* (1992), table 1. From a Monte Carlo simulation of 4000 values of KL_{mn} , it is found that

$$P(\widehat{KL}_{mn} \leq 0.9146 | F_0 = \text{Gamma}(4, 1)) = 0.8850$$

This gives a good indication that the sample could be drawn from a Gamma $(k = 4, 1)$ distribution.

3. A VARIATION ON THE KL MEASURE

Several other measures of discrimination between F and F_0 exist. The following statistics are a few to estimate such measures given the sample $x_{(1)}, \dots, x_{(m)}$:

$$\text{Van Soest (1969):} \quad \sum_1^n \left\{ F_0(x_{(i)}) - \frac{2i-1}{n} \right\}^2 + \frac{n}{12}$$

$$\text{Finkelstein and Schafer (1971):} \quad \sum_1^n \max \left\{ \left| F_0(x_{(i)}) - \frac{i}{n} \right|, \left| F_0(x_{(i)}) - \frac{i-1}{n} \right| \right\}$$

$$\begin{aligned} \text{Bickel and Docksum (1977):} \quad & \text{(i) } \sum_1^n \left| F_0(x_i) - \frac{i}{n} \right|, \\ & \text{(ii) } \max \left| F_0(x_i) - \frac{i}{n} \right|. \end{aligned}$$

The measure we are proposing and which is somewhat related to the KL measure is

$$(6) \quad J(F, F_0) = \int_0^w f(x) \ell n \frac{F(x)}{F_0(x)} dx.$$

$$\begin{aligned} \text{Since} \quad J(F, F_0) &= \int_0^w f(x) \ell n F(x) dx - \int_0^w f(x) \ell n F_0(x) dx \\ &\geq \int_0^w f(x) \ell n F(x) dx \\ &= -1, \end{aligned}$$

it follows that $-1 < J(F, F_0) < \infty$ for all F and F_0 defined over a non-negative support.

It can be shown that for a Gamma $(k, 1)$ distribution $J(F, F_0) = 0$ iff $k = k_0$.

(Personal communication with Ph.D. student J.H. Meyer, UOFS.)

A typical graph of $J(F, F_0)$ is shown in figure 1 for $k_0 = 4$ and $1 \leq k \leq 10$.

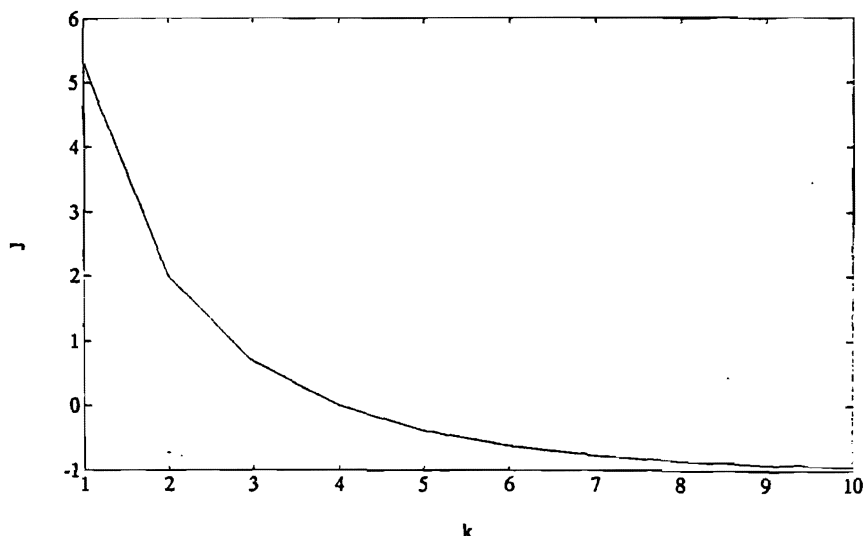


FIGURE 1. $J(F, F_0)$ plotted against k for $k_0 = 4$

The following estimate of $J(F, F_0)$ is considered, given the sample $X_1, X_2, \dots, X_n$:

$$(7) \quad \hat{J}_n = \frac{1}{n} \sum_{i=1}^n \ell n \left\{ \frac{F_n(X_i)}{F_0(X_i)} \right\}$$

where $F_n(X_i)$ is an empirical estimate of $F(X_i)$. The unbiased estimate

$F_n(x) = \frac{1}{n}(\#X_i \leq x)$ is a good one to use. The asymptotic distribution of $\hat{J}_n$ is, however, not difficult to derive and is given in the following theorem:

THEOREM: If $F_0(x)$ is specified to be a Gamma $(k_0, 1)$ distribution function, then given a random sample $X_1, \dots, X_n$ from F_0 the estimator $\hat{J}_n = \frac{1}{n} \sum_{i=1}^n \ell_n \left\{ \frac{F_n(X_i)}{F_0(X_i)} \right\}$ is asymptotically normally distributed with mean 0

and variance

$$(8) \quad V(\hat{J}_n) = \frac{1 - F_0(k_0)}{n^2 F_0(k_0)} + \frac{f_0(k_0)}{2n^2 F_0^2(k_0)} + \frac{k_0 f_0^2(k_0)}{n^2 F_0^3(k_0)}$$

where $F_0(k_0) = \int_0^{k_0} f_0(x) dx$, $f_0(x) = \frac{1}{\Gamma(k_0)} e^{-x} x^{k_0-1}$.

PROOF: It is known (Rice, 1988, page 315) that

$$E\{F_n(x)\} = F(x) \quad \text{and} \quad V\{F_n(x)\} = \frac{1}{n} F(x)(1 - F(x)).$$

Hence, using the delta method,

$$E\{\ell_n F_n(x)\} \approx \ell_n F(x)$$

$$V\{\ell_n F_n(x)\} \approx \frac{1}{n} \frac{1 - F(x)}{F(x)}$$

For X random, $E\{\ell_n F_n(X)\} \approx E \ell_n F(X)$ and $V\{\ell_n F_n(X)\} \approx \frac{1}{n} E\left\{\frac{1 - F(X)}{F(X)}\right\} + V\{\ell_n F(X)\}$.

Therefore

$$E(\hat{J}_n) = 0$$

$$\begin{aligned} V(\hat{J}_n) &= \frac{1}{n^2} \sum_{i=1}^n V\left\{\log \left[\frac{F_n(X_i)}{F_0(X_i)}\right]\right\} \\ &\approx \frac{1}{n^2} \sum_{i=1}^n E\left\{\frac{1 - F_0(X_i)}{F_0(X_i)}\right\}/n \\ &\approx \frac{1 - F_0(k_0)}{n^2 F_0(k_0)} + \frac{f_0(k_0)}{2n^2 F_0^2(k_0)} + \frac{k_0 f_0^2(k_0)}{n^2 F_0^3(k_0)}. \end{aligned}$$

$$\text{Since } E(X_i) = k, \quad F_0(k_0) = \frac{1}{\Gamma(k_0)} \int_0^{k_0} e^{-x} x^{k_0-1} dx,$$

$$f_0'(k_0) \frac{d}{dx} f_0(x) \Big|_{x=k_0} = -\frac{f_0(k_0)}{k_0}, \quad f_0(k_0) = \frac{1}{\Gamma(k_0)} e^{-k_0} k_0^{k_0-1}.$$

This variance is an approximation and under-estimates the true variance, but serves to judge on selecting k_0 . The next example serves as an illustration.

Example 2

For the same data considered in example 1, we found $\hat{J}_n = 0.0379$ given $k_0 = 4$. The variance according to theorem 1 is calculated as $V(\hat{J}_n) = 0.0209^2$. There is therefore no reason to look for another k_0 value and we conclude that the sample could be drawn from a Gamma (4, 1) distribution. This coincides with the result of example 1.

Another advantage of using $J(F, F_0)$ as a goodness of fit measure instead of $KL(F, F_0)$ is that negative values for $J(F, F_0)$ indicate that k_0 is larger than the true k and if $J(F, F_0)$ is positive k_0 is smaller than the true k . See figure 1.

4. SELECTING THE APPROPRIATE MODEL FROM THE CLASS OF GENERALIZED EXTREME VALUE DISTRIBUTIONS

We will now consider the class of Generalized Extreme Value distributions and will describe a procedure to select the k value that will give the best fit among the class of GEV distributions, that is selecting the appropriate model, that will fit the data. In de Waal *et al.* (1995), the estimation of k is considered using a Jeffreys' prior or the reference prior for k .

The Gibbs sampler is then used for estimating the parameters in a linear model under the GEV distribution. This procedure works well although it

needs much computation. The approach suggested here is quick and serves as an alternative for selecting the model. It may be preferred as a way of testing the goodness of fit using the KL measure I or this variation on the KL measure, J , for fitting a GEV distribution to data.

If the random variable Y is distributed $\text{GEV}(k, \mu, \sigma^2)$ with mean μ and variance σ^2 , then the density function of Y is given by (de Waal *et al.*, 1995)

$$(9) \quad f(y) = \frac{\sqrt{\psi'(k)}}{\Gamma(k)} \frac{1}{\sigma} \exp(-v_y) (v_y)^k, \quad -\infty < y < \infty$$

where

$$v_y = \exp\left\{\psi(k) - \frac{(y - \mu)}{\sigma} \sqrt{\psi'(k)}\right\},$$

$$\psi(k) = \frac{d}{dk} \ln \Gamma(k), \quad \psi'(k) = \frac{d}{dk} \psi(k).$$

It can be shown that $V_y = \exp\left\{\psi(k) - \frac{(Y - \mu)}{\sigma} \sqrt{\psi'(k)}\right\}$ is distributed Gamma $(k, 1)$ with density function

$$(10) \quad g(v_y) = \frac{1}{\Gamma(k)} e^{-v_y} v_y^{k-1}, \quad v_y > 0.$$

Since we mainly look at lifetime variables where the lognormal, gamma, Weibull and exponential distributions are of interest, we can show that if the lifetime random variable T follows the Generalized Gamma (GGamma) distribution with parameters k, μ and σ and density function given by

$$(11) \quad h(t) = \frac{\sqrt{\psi'(k)}}{\Gamma(k)} \cdot \frac{1}{\sigma} \cdot \frac{1}{t} e^{-v_t} v_t^k, \quad t > 0$$

where

$$v_t = \exp\left\{\psi(k) - \frac{(-\ln t - \mu)}{\sigma} \sqrt{\psi'(k)}\right\},$$

then $-\ln T$ follows a $\text{GEV}(k, \mu, \sigma^2)$ distribution.

Therefore, by fitting a GEV distribution from data $y_1, y_2, \dots, y_n$ on the random variable Y , we are also fitting a GGamma distribution from data $t_1, t_2, \dots, t_n$ to the lifetime variable T .

Estimates of μ and σ are

$$\hat{\mu} = \bar{y} = \frac{1}{n} \sum y_i, \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum (y_i - \bar{y})^2$$

respectively. The selection of k can be found by estimating J (or I) described in section 3 (or 2), since V_y is distributed Gamma ($k, 1$). Hence the cdf of Y, F_y , can be written as

$$\begin{aligned}
 (12) \quad P(Y < y) &= P(V_y > v_y) \\
 &= 1 - \Gamma(k, v_y) \\
 &= 1 - \frac{1}{\Gamma(k)} \int_0^{v_y} e^{-x} x^{k-1} dx \\
 &= 1 - F(t).
 \end{aligned}$$

The empirical estimate of $F(t)$, $F_n(t)$, can then be compared with $F_0(t)$ by calculating $\hat{J}_n$ and then selecting k_0 for which $\hat{J}_n$ is closest to zero.

Let us consider a few examples.

Example 3

Grubbs (1971) recorded the mileages t for 19 military personnel carriers that failed in service. The mileages are

$t:$	162	200	271	230	393	508	539	629
	706	778	884	1003	1101	1182	1463	1603
	1984	2355	2880					

Transforming the t -values to $y = -\ln t$, we got the estimates for μ and σ as $\hat{\mu} = -6.6145$ and $\hat{\sigma} = 0.8208$. Selecting a few k_0 values, the following estimates for $J(F_1 F_0)$ were obtained:

$$\begin{aligned}
 k_0 = 4 & : \hat{J} = 0.1223 \\
 k_0 = 5 & : \hat{J} = 0.1200 \\
 k_0 = 10 & : \hat{J} = 0.1172 \\
 k_0 = 20 & : \hat{J} = 0.1164
 \end{aligned}$$

The values of $\hat{J}$ are changing very slowly as k_0 increases, indicating that a GEV distribution with $k_0 = 10$ or $k_0 = 20$, for instance, may fit. In fact, fitting a $N(\hat{\mu}, \hat{\sigma}^2)$ distribution where $k_0 \rightarrow \infty$, shows almost the same fit as a GEV with $k_0 = 20$ (see figures 2, 3 and 4).

The value of $\hat{KL}_{mn}$ for $m = 12$ is 0.0353 if k_0 is chosen equal to 10, also converging slowly to zero as $k_0 \rightarrow \infty$ (p value = 0.105).

These fits are, however, better than the exponential fit proposed by Ebrahimi *et al.* (1992) where $\hat{J} = 0.1451$ (see figure 4).

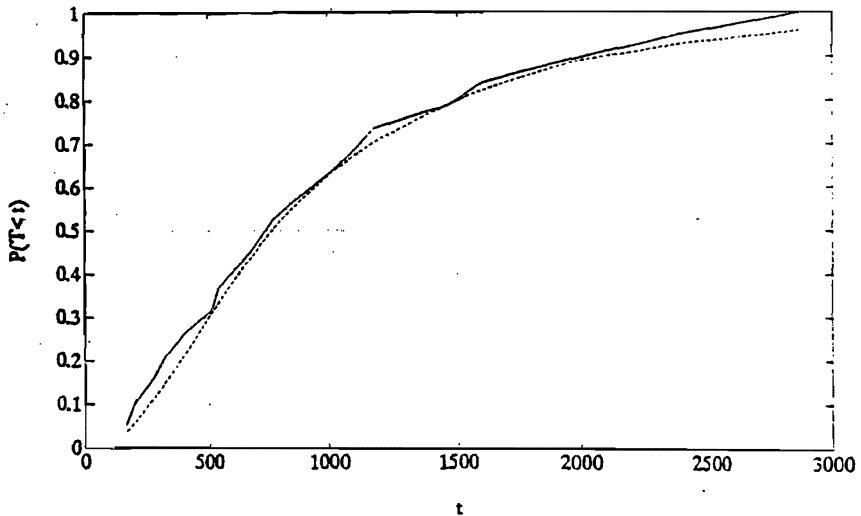


FIGURE 2. Empirical cdf (—) compared with the cdf of the GEV distribution (---) with $k_0 = 20$, from the Grubbs data

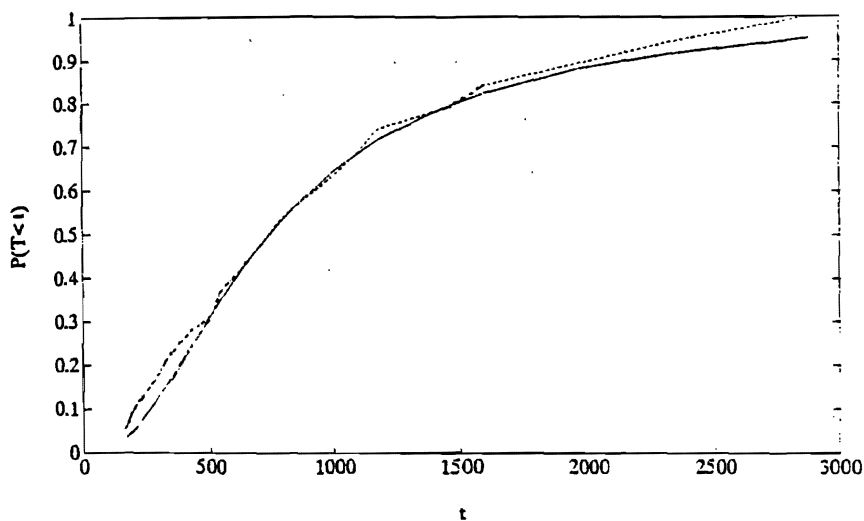


FIGURE 3. Empirical cdf (---) compared with the cdf of a Normal (—) from the Grubbs data

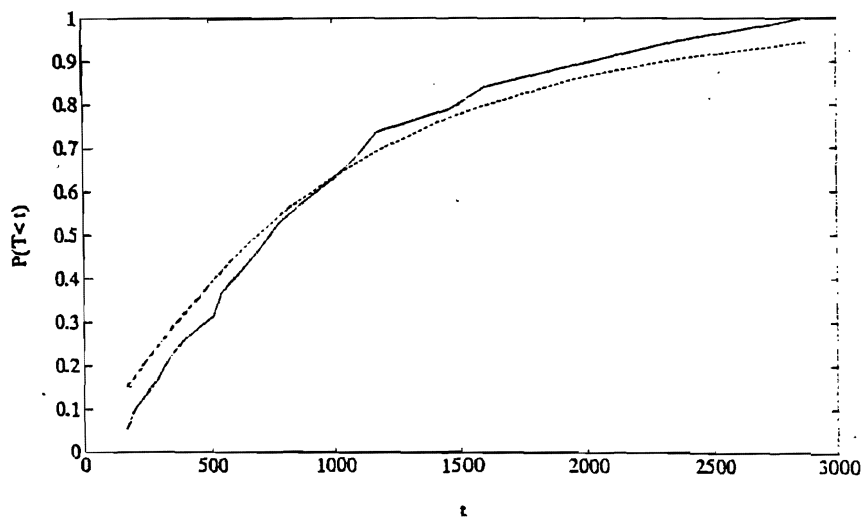


FIGURE 4. Empirical cdf (—) compared with the cdf of the Exponential distribution (---) from the Grubbs data

Example 4

In de Waal *et al.* (1995), data are given on the time three Farr yachts sailed 66 races with different wind speeds and different lengths of courses. The time the yacht sailed the course is the random variable T and the wind speed and length of course are considered covariates X_1 and X_2 . A linear model is assumed of $Y = -\ln T$ on X_1 and X_2 . A GEV ($k, \mu = \beta_0 + \beta_1 X_1 + \beta_2 X_2, \sigma^2$) model is assumed and the following estimates for $k_0 = 4$ of the parameters β_0, β_1 , and β_2 obtained:

$$\hat{\beta}_0 = -0.09194, \hat{\beta}_1 = 0.0436, \hat{\beta}_2 = -0.0507, \hat{\sigma}^2 = 0.2181^2.$$

The errors $e_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{1i} - \hat{\beta}_2 x_{2i}$ are given in table 1.

TABLE 1. Values of $e = y - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \hat{\beta}_2 x_2$

-0.4576	-0.1869	-0.1013	-0.0451	0.0769	0.2305
-0.4536	-0.1783	-0.0997	-0.0274	0.0775	0.2394
-0.3735	-0.1785	-0.0979	-0.0205	0.0839	0.2716
-0.2992	-0.1741	-0.0891	-0.0020	0.0871	0.2755
-0.2470	-0.1705	-0.0857	-0.0013	0.0891	0.3393
-0.2421	-0.1586	-0.0852	0.0044	0.0915	0.4036
-0.2314	-0.1473	-0.0849	0.0292	0.1086	0.4062
-0.2259	-0.1454	-0.0666	0.0305	0.1393	0.4134
-0.2003	-0.1230	-0.0585	0.0310	0.2181	0.4469
-0.1950	-0.1036	-0.0552	0.0526	0.2228	0.4755
-0.1936	-0.1015	-0.0452	0.0681	0.2289	0.8119

To judge if the model is good for predicting the time one of those yachts will sail a course of specified length under certain wind conditions, we can fit a GEV distribution with $k_0 = 4$, $\mu = 0$ and $\hat{\sigma}^2 = 0.2181^2$ to these errors. This has been done and $\hat{J}_{n=66}$ calculated as $\hat{J}_{66} = 0.0514$. This estimate is

quite close to zero, considering the fact that $V(\hat{J}_{66}) = 0.1205^2$, so we can be satisfied that the GEV distribution with $k_0 = 4$ and the estimates of β_0 , β_1 and β_2 given above if $k_0 = 4$ is quite an appropriate model to use. The fit is shown in figure 5.

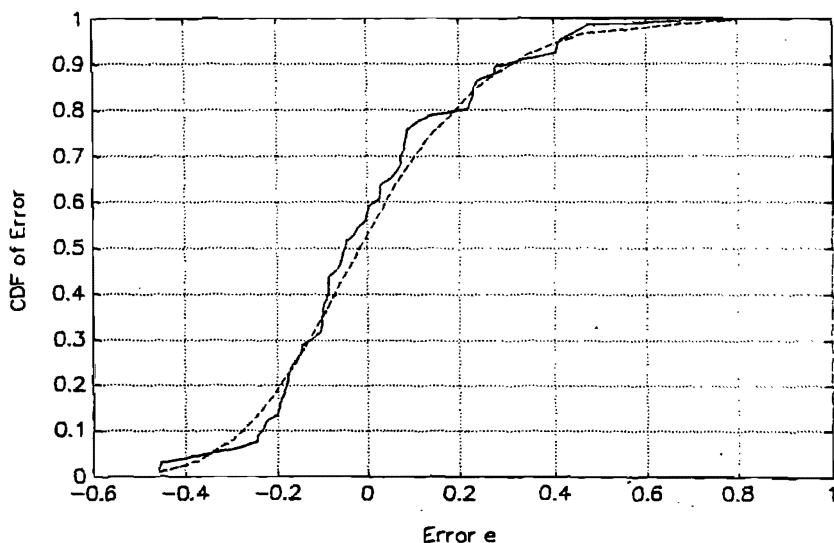


FIGURE 5. Empirical cdf of errors (—) and cdf of GEV distribution with $k_0 = 4$ for the Farr data

The estimates of β_0 and σ^2 can be adjusted to bring the mean of e closer to zero and the estimate of σ^2 closer to the calculated variance of the errors. In this way one can find the best fitted GEV distribution, that is where $\hat{J}$ is close to zero. It turns out that for the above case that $k = 10$ gives a good fit, which implies that the normal model may also be appropriate to use.

5. CONCLUSION

We have demonstrated that a suitable distribution among the class of Generalized Extreme Value distributions can be selected fairly easily by using a variation of the Kullback-Leibler measure of divergence between two distributions. We have also shown that the fitting of a GEV (k, μ, σ^2) distribution is quick and easy with the aid of the facility to compute an incomplete Gamma integral. In lifetime data it quite often becomes a question of which distribution gives the best fit. The class of GEV distributions will take care of this and knowledge of the use of this class becomes useful.

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